# Paper Solution

# EGD – Winter 2022

## Engineering Graphics and Design (GTU)

**Q.1 (a)** Define representative fraction (R.F.) of scale. Also list types of scale according to value of R.F. **( 03 Marks)**

**Answer:**

Representative Fraction is the ratio of size of object in paper to actual size of object.

$$\text{Representative Fraction (R.F.)} = \frac{\text{Size of object in paper}}{\text{Actual size of object}}$$

According to value of R.F., we can explain full scale, reduced scale and enlarged scale.

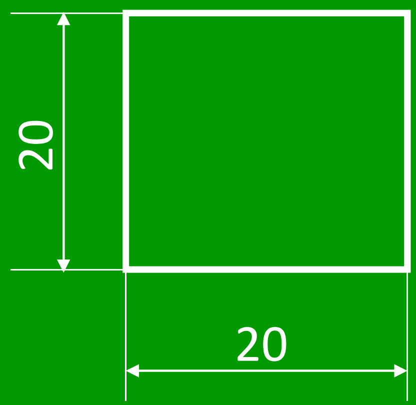
**Full Scale:**

Draw 20mm size square.

$$\text{Representative Fraction (R.F.)} = \frac{\text{Size of object in paper}}{\text{Actual size of object}}$$

$$\text{Representative Fraction (R.F.)} = \frac{20\text{mm}}{20\text{mm}} = \frac{1}{1} = 1:1$$

Here, size of object in paper and actual size of object is same.




Scale 1:1

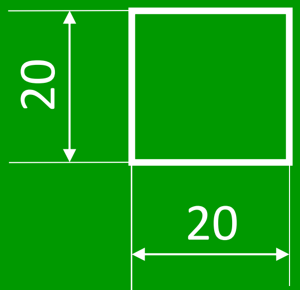
**Use:**
It is used in general purpose drawing.

**Reduced Scale:**

Draw 20mm size square.

Representative Fraction (R.F.) = $\dfrac{\text{Size of object in paper}}{\text{Actual size of object}}$

Representative Fraction (R.F.) = $\dfrac{10\text{mm}}{20\text{mm}} = \dfrac{1}{2} = 1:2$




Scale 1:2

Here, size of object in paper is less than actual size of object.

**Other Example:**

1:5, 1:10, 1:20, 1:50, 1:100, 1:1000, 1:100000

**Use:**

When size of object is too large.

Example: Building drawing, Machinery drawing, map etc.

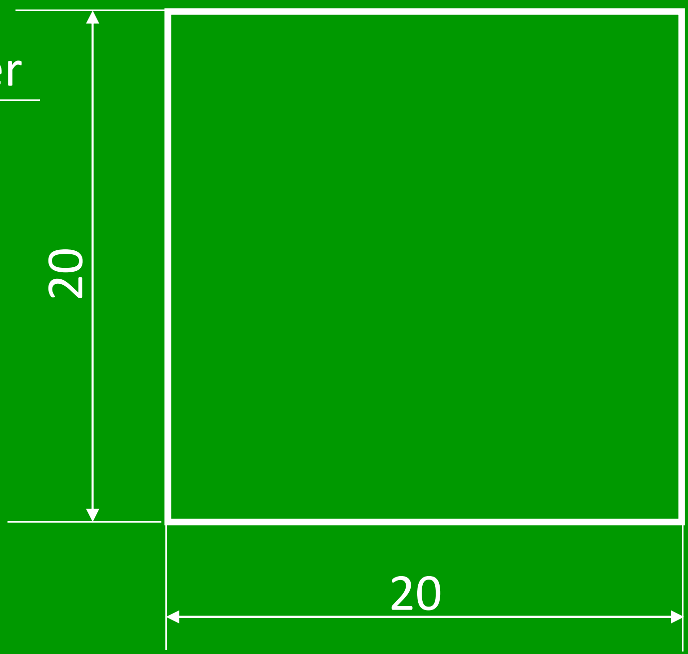
**Enlarged Scale:**

Draw 20mm size square.

$$\text{Representative Fraction (R.F.)} = \frac{\text{Size of object in paper}}{\text{Actual size of object}}$$

$$\text{Representative Fraction (R.F.)} = \frac{40\text{mm}}{20\text{mm}} = \frac{2}{1} = 2:1$$

Here, size of object in paper is greater than actual size of object.




Scale 2:1

**Other Example:**

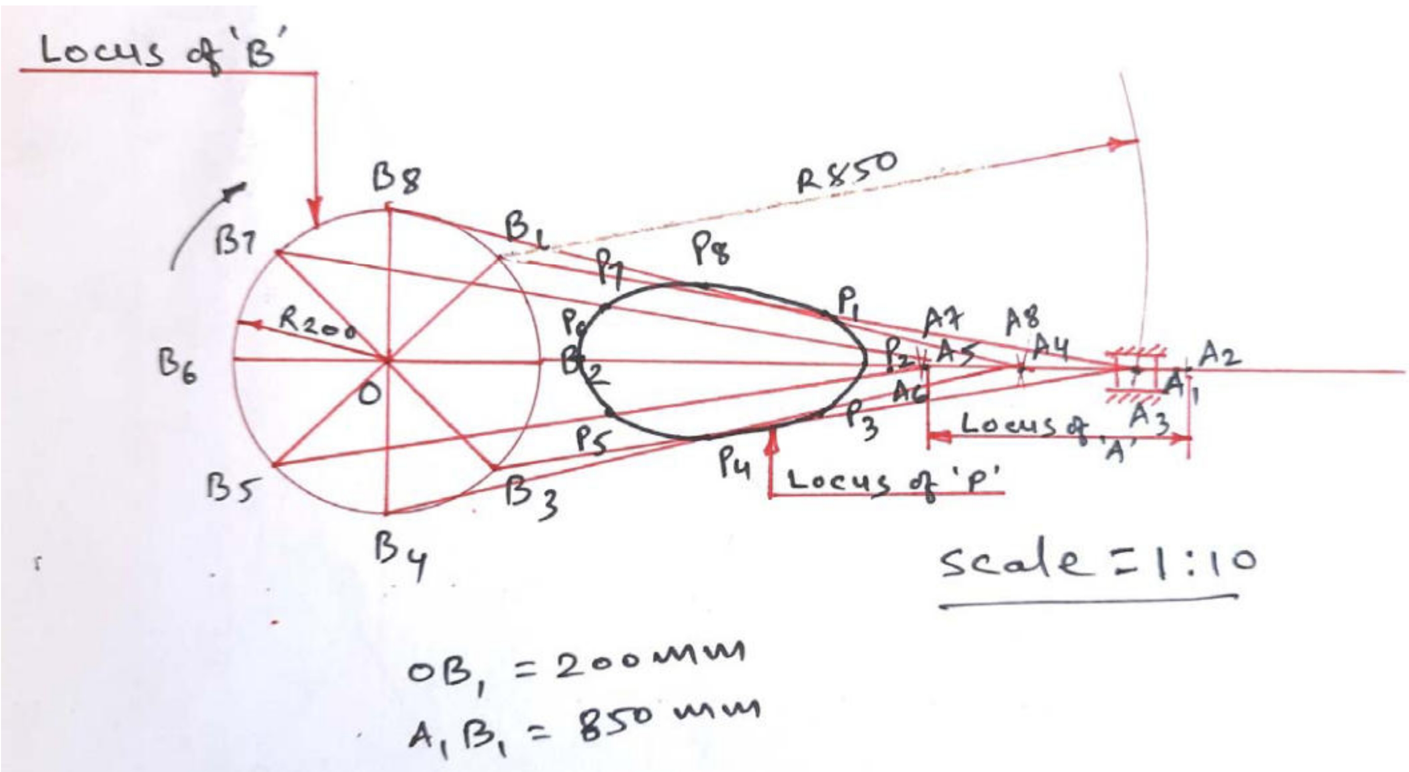
5:1, 10:1, 20:1, 50:1 etc.

**Use:**

When size of object is too small.

Example: Parts of wrist watch, parts of electronic circuit etc.

**Q.1 (b)** In a slider crank mechanism OBA, the crank OB is 200 mm long and the connecting rod BA is 850 mm long. Plot the locus of point P where P is midpoint of connecting rod. Refer figure 1. **( 04 Marks)**



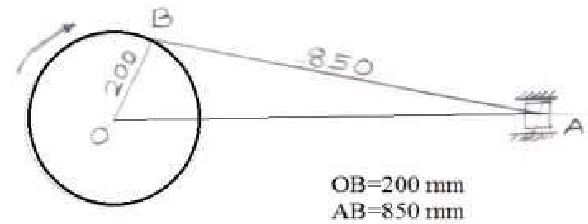
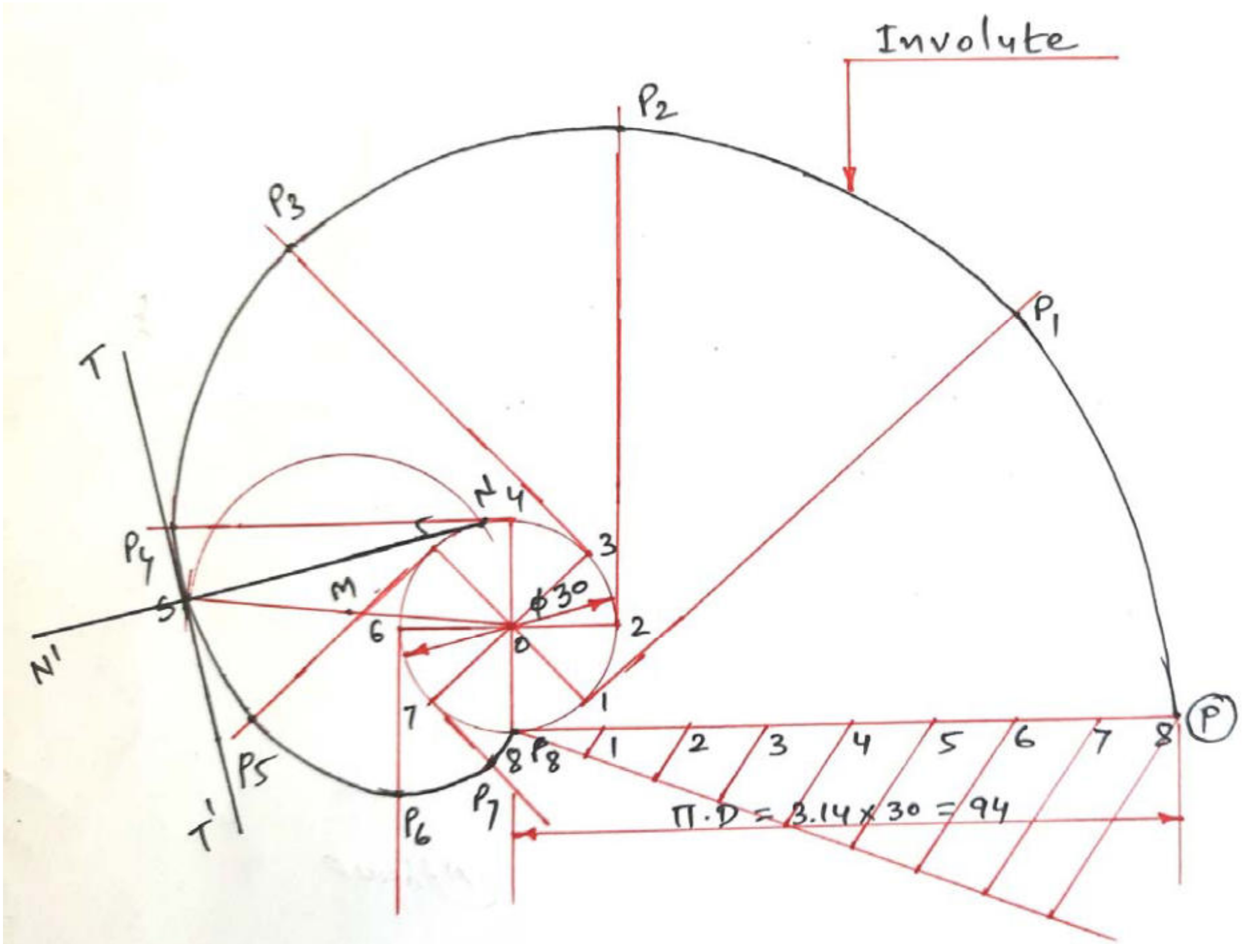
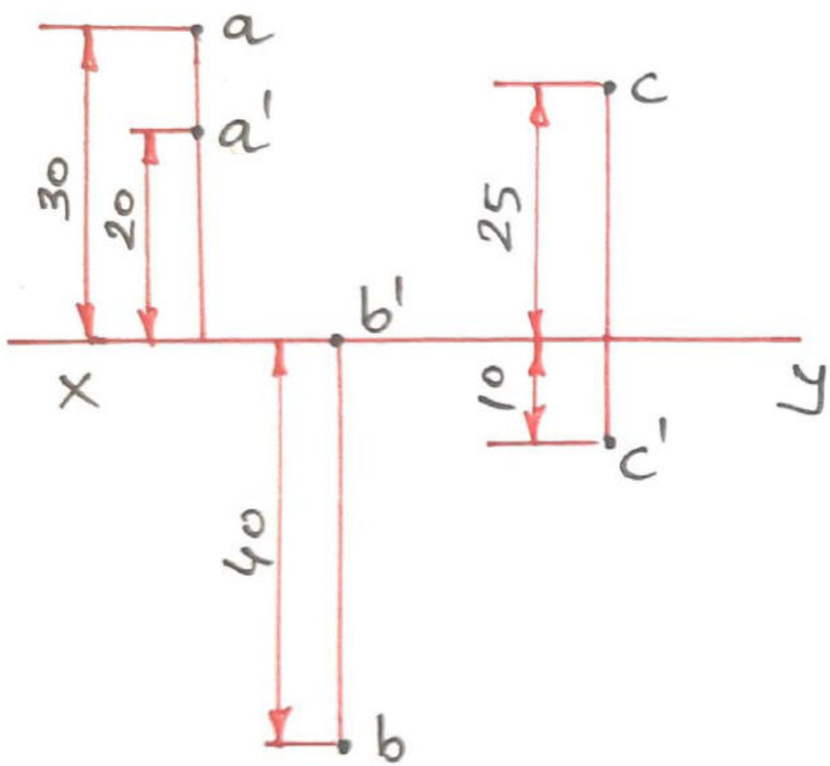




Figure 1

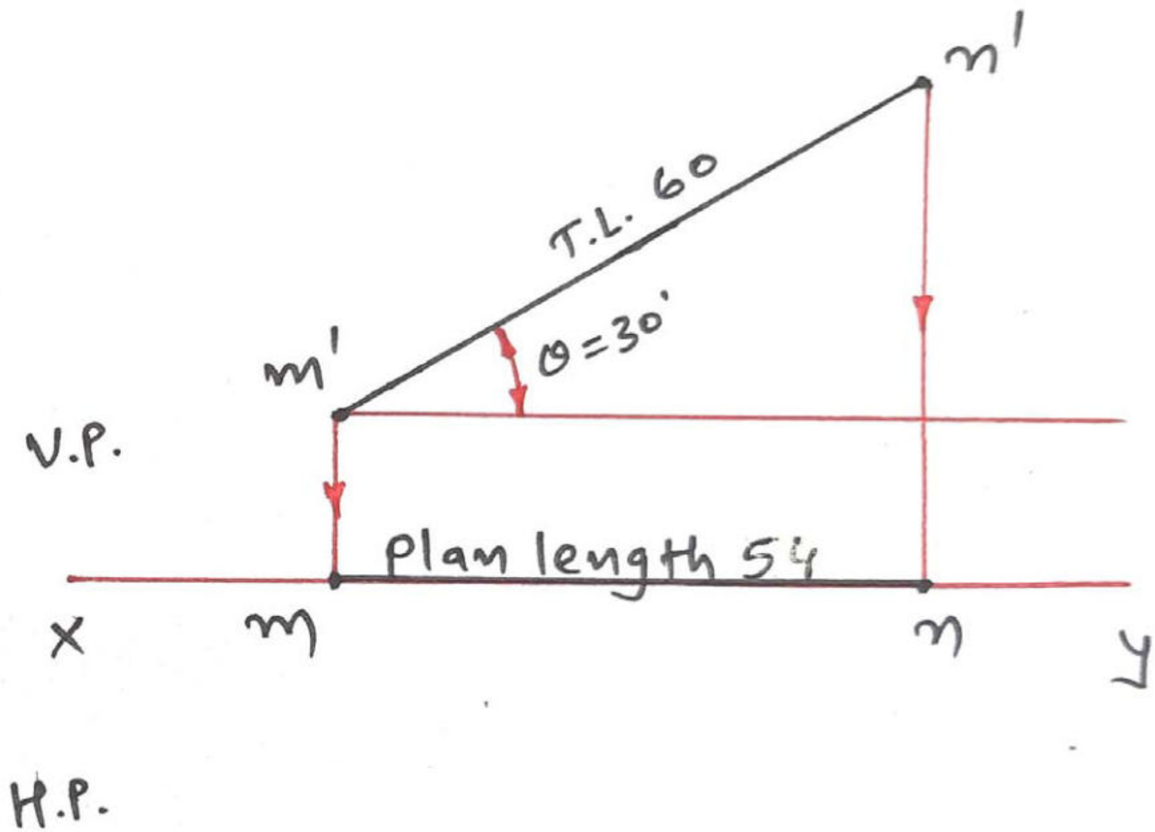
**Q. 1 (c)** Construct one complete turn of an involute of a circle having diameter 30 mm. Also draw tangent and normal to the curve S. The point S is a point on the involute and it is at a distance of 45mm from centre of the circle. **( 07 Marks)**



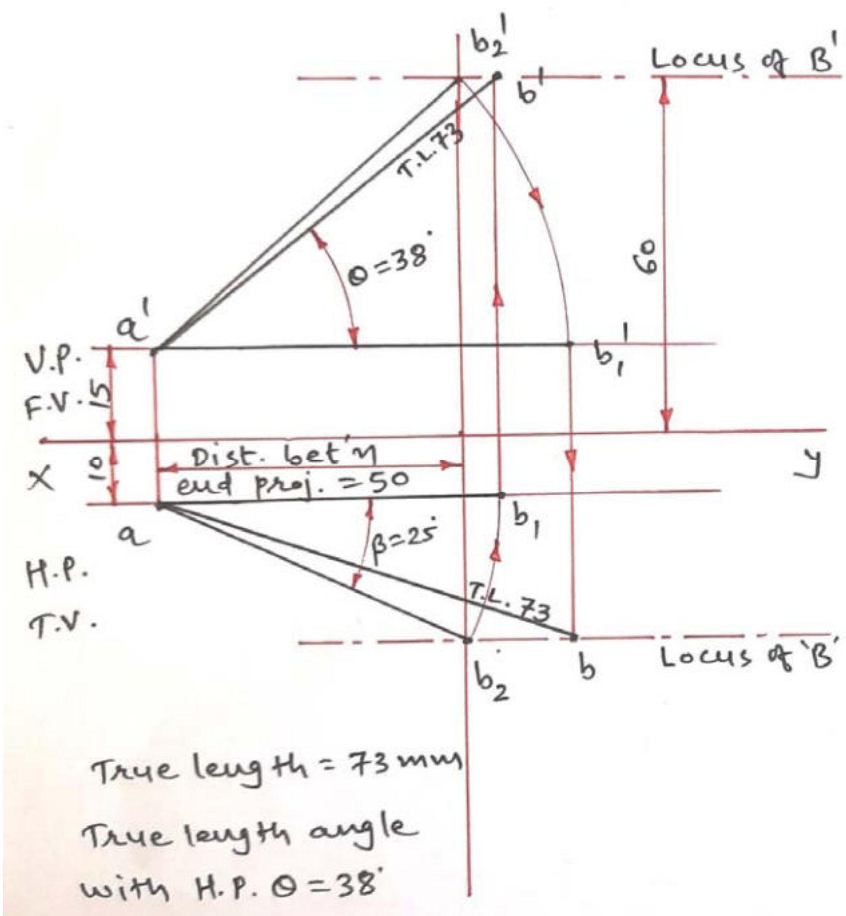
**Q.2 (a)** Draw the projection of following points on the same X-Y line.
(1) Point A is 20 mm above HP and 30 mm behind V.P.
(2) Point B is on HP and 40 mm in front of V.P.
(3) Point C is 10 mm bellow HP and 25 mm behind V.P. **(03 Marks)**



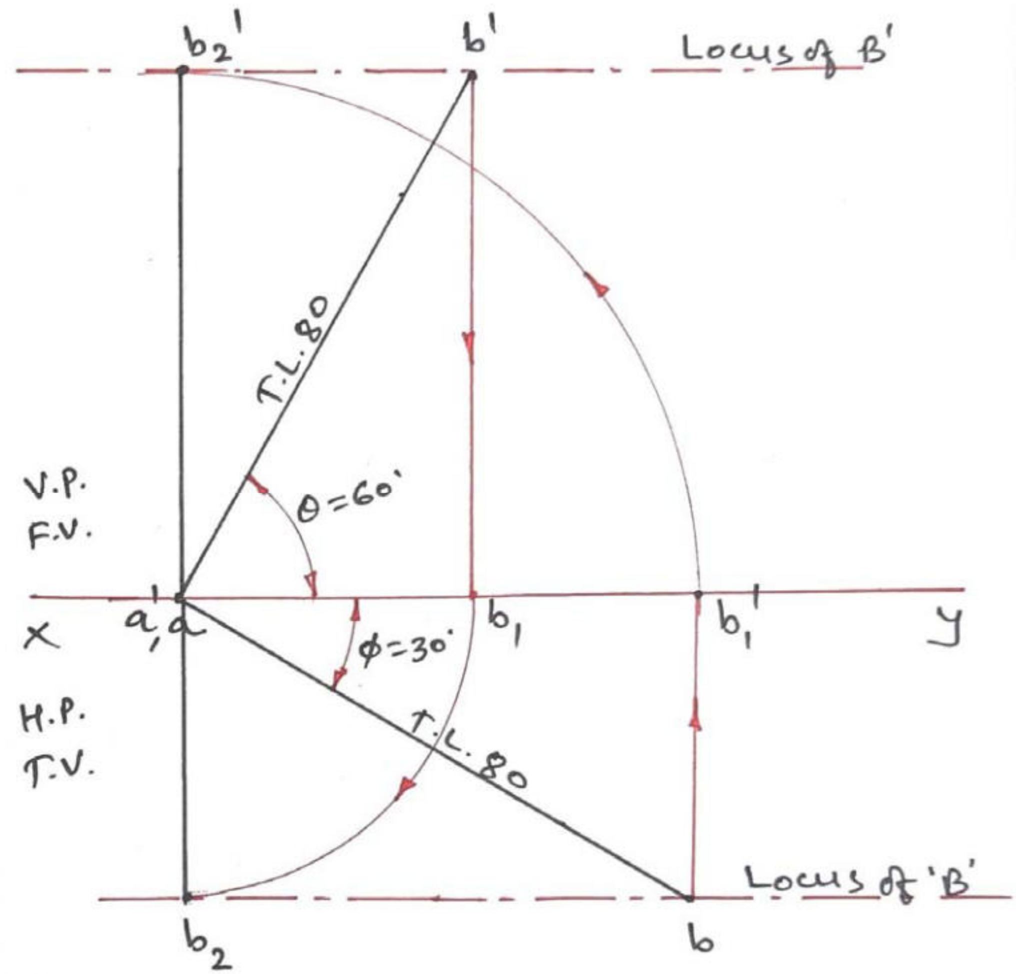
**Q. 2 (b)** A line MN, 60 mm long, is in V.P. It makes an angle of 30° with the H.P. The end point M is 15 mm above H.P. Draw projections of line MN. Also measure Plan length of the line. **( 04 Marks)**



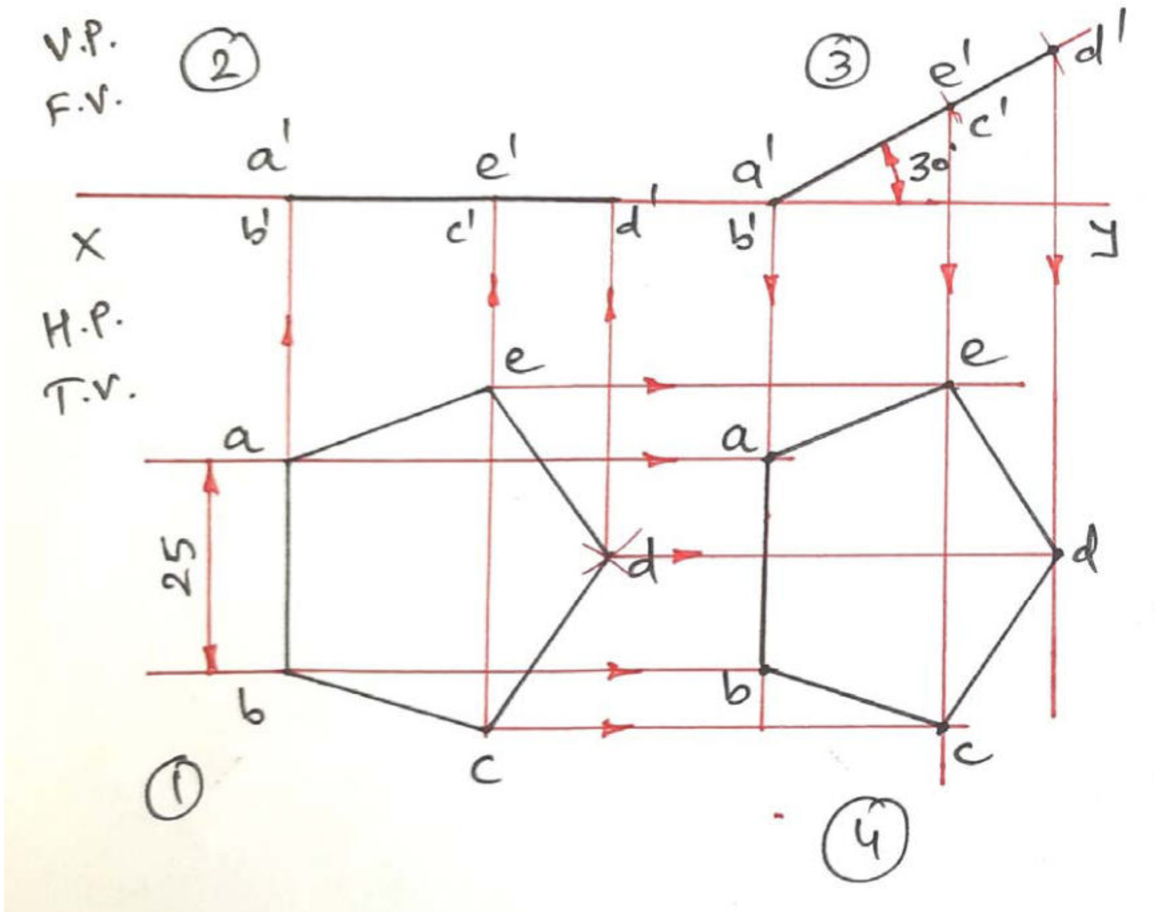
**Q. 2 (c)** The line AB has its end A, 15 mm above H.P. and 10 mm in front of V.P. The end B is 60 mm above H.P. The distance between the end projectors is 50 mm. The plan of the line is inclined at 25° to X-Y line. Draw the projections of line. Find true length and inclination of the line with H.P. **( 07 Marks)**



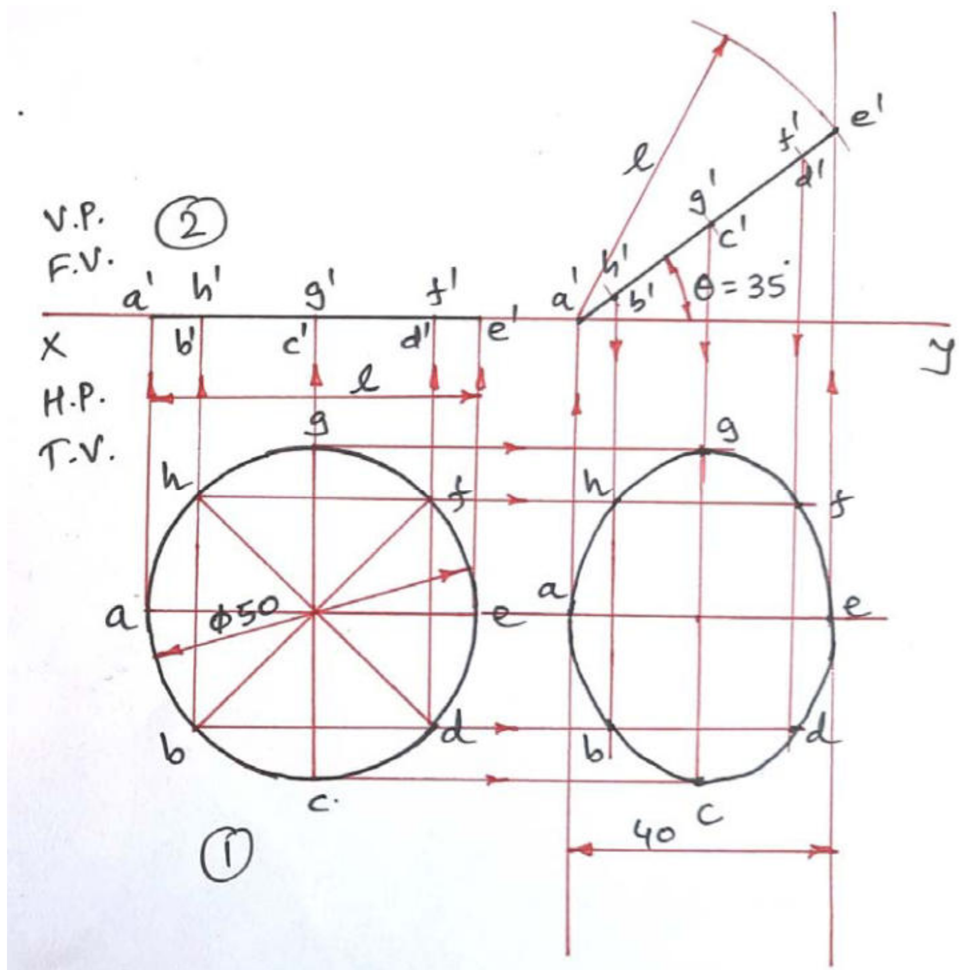
**Q. 2 (c) OR** Draw the projections of a straight line 80 mm long inclined at 60° to H.P. and 30° to V.P. with end A in the H.P. and the end B in the V.P. **( 07 Marks)**



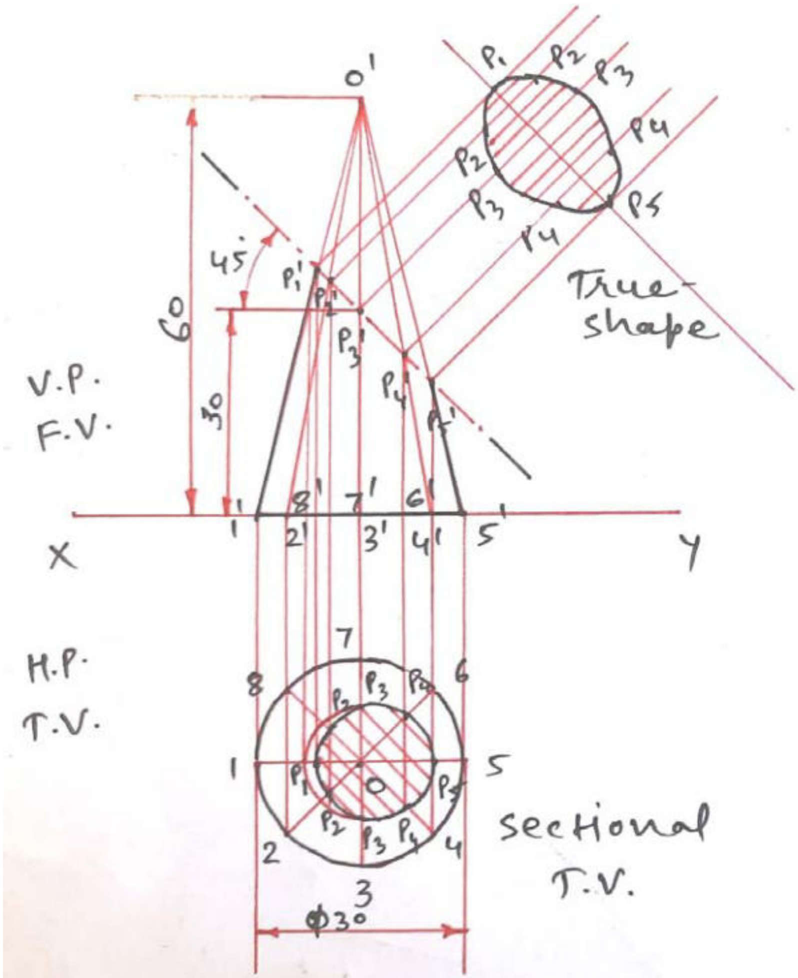
**Q.3 (a)** A pentagonal plate, side 25 mm, is inclined to H.P. by 30°.It is perpendicular to V.P. and resting on HP on its one of the side. Draw projections of the plate. **( 03 Marks)**



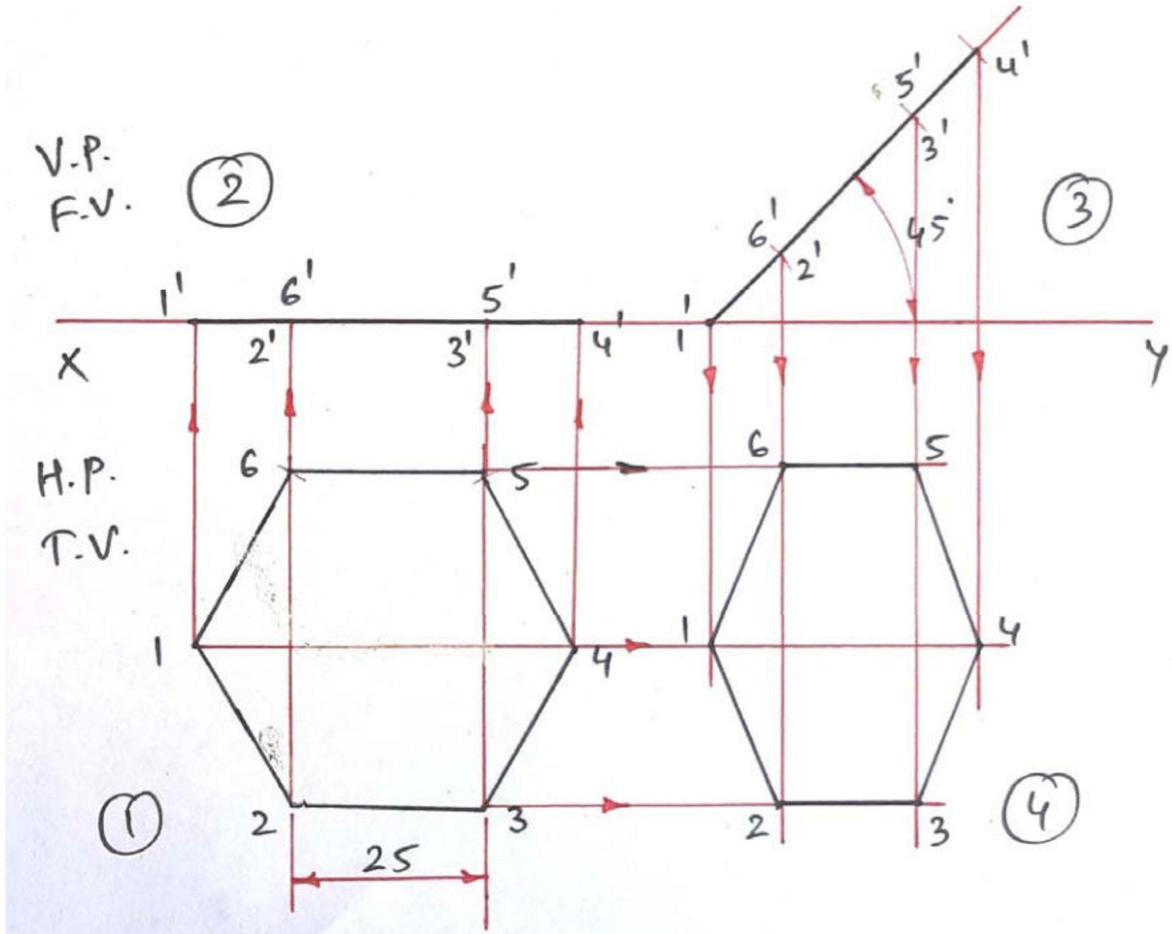
**Q. 3 (b)** A circular disk, 50 mm in diameter, is resting on the H.P. on one of the point A of the circumference. Plane is inclined to the H.P. such a way that the top view of it is an ellipse of minor axis 40 mm. Draw its projections. **( 04 Marks)**



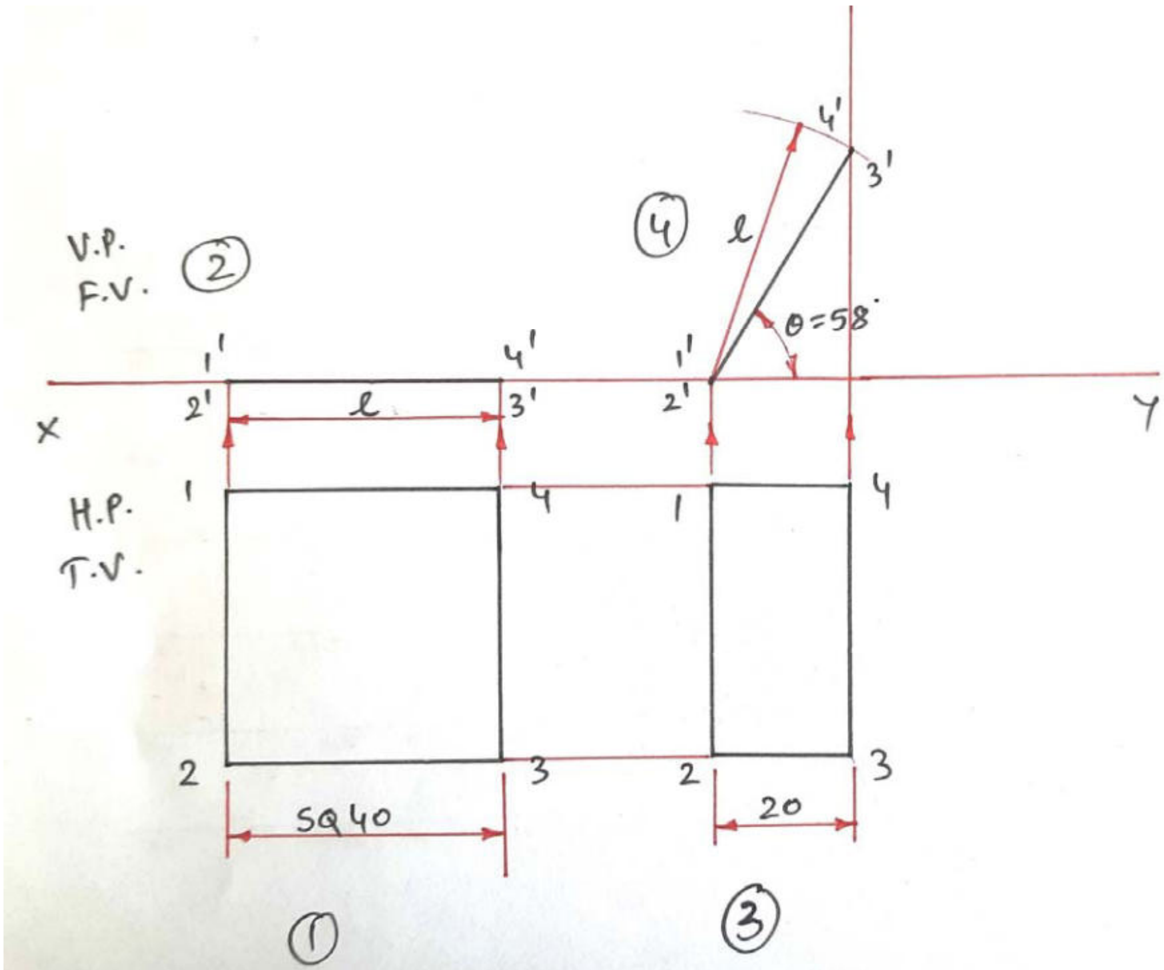
**Q. 3 (c)** A cone, base diameter 30 mm and height 60 mm, is resting on its base. It is cut by AIP inclined at 45°to H.P. and passing through midpoint of the axis of cone. Draw front view, sectional top view and true shape of this sectional solid. **( 07 Marks)**



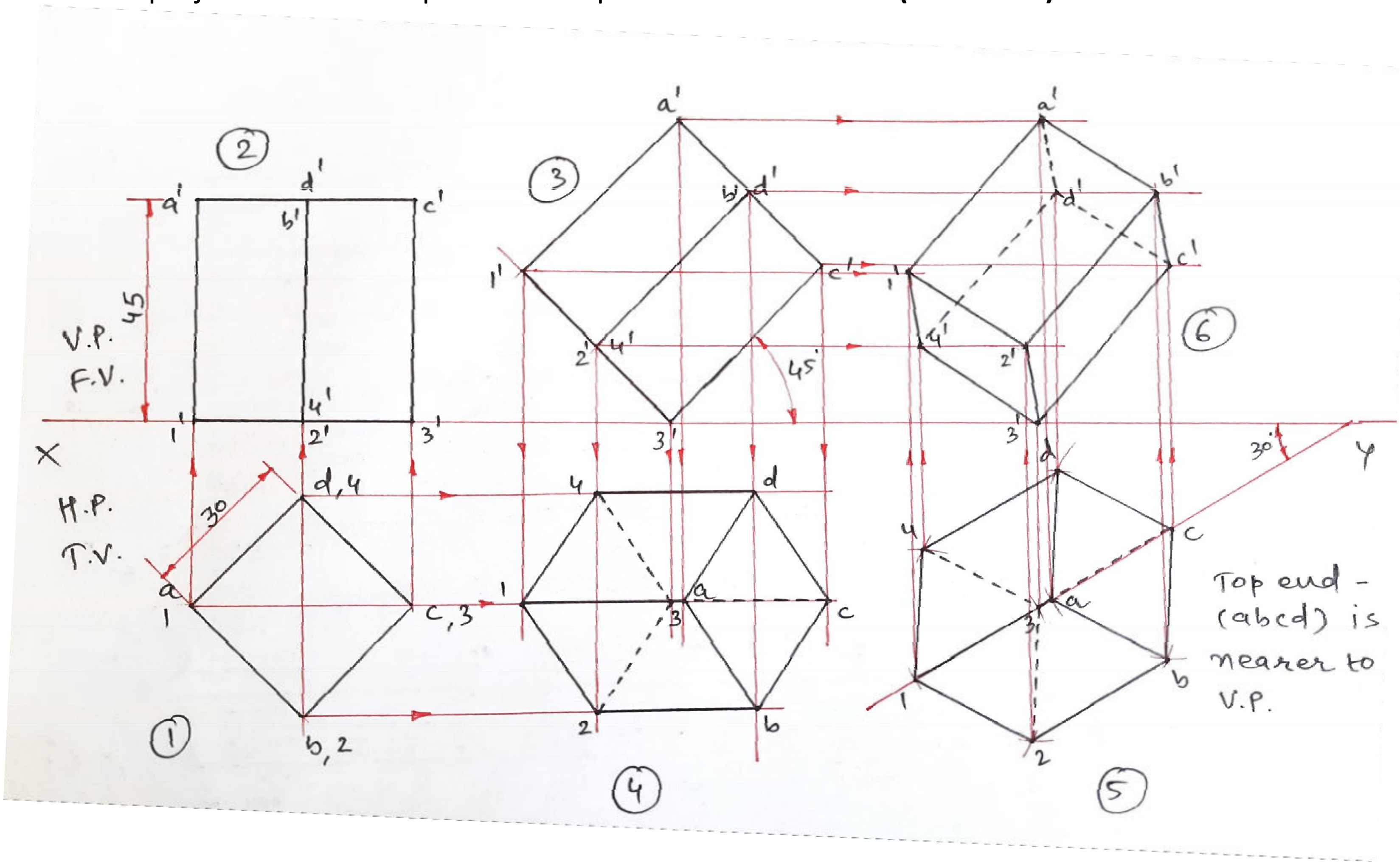
**Q.3 (a) OR** A hexagonal plate, side 25 mm, is inclined to H.P. by 45°. It is resting on HP on one of the corner. The plate is perpendicular to V.P. Draw projections of the plate. **( 03 Marks)**



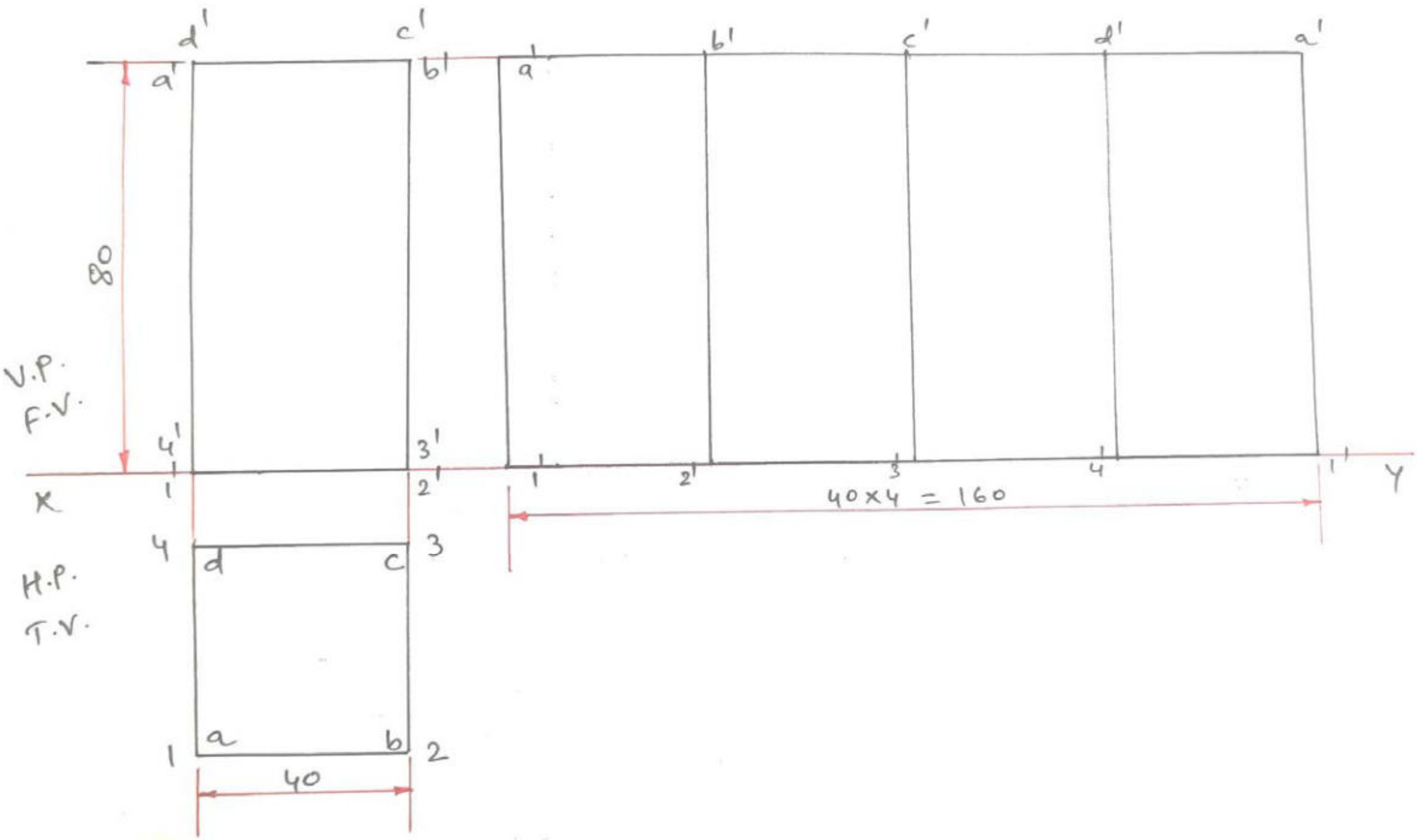
**Q. 3 (b) OR** A square plate, 40 mm side, is resting on the H.P. on one of its sides. The plate is inclined to H.P such a way that it's top view looks like a rectangle with 20 mm shorter side. Draw its projections. **( 04 Marks)**



**Q. 3 (c) OR** A square prism, side of base 30 mm and height 45 mm, is resting on H.P. on one of the corners of the base. The longer edge containing that corner on which it rests, is inclined at 45° to H.P. and top view of it is inclined at 30° to V.P. Draw the projections when top end of the prism is nearer to V.P. **( 07 Marks)**



**Q.4 (a)** Develop the complete surface of a square prism of side of base 40 mm and height 80 mm. **( 03 Marks)**



**Q.4 (b)** Draw front view of an object shown in figure 2 using first angle projection methods. **( 04 Marks)**

**Q. 4 (c)** Draw top view and left hand side view of object shown in figure 2 using first angle projection methods. **( 07 Marks)**

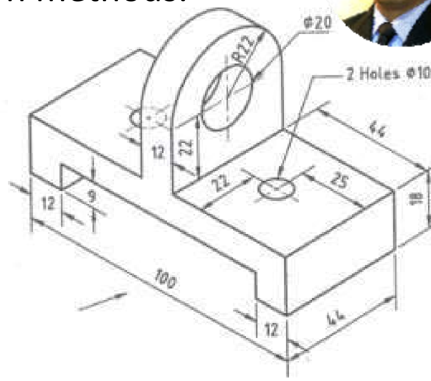
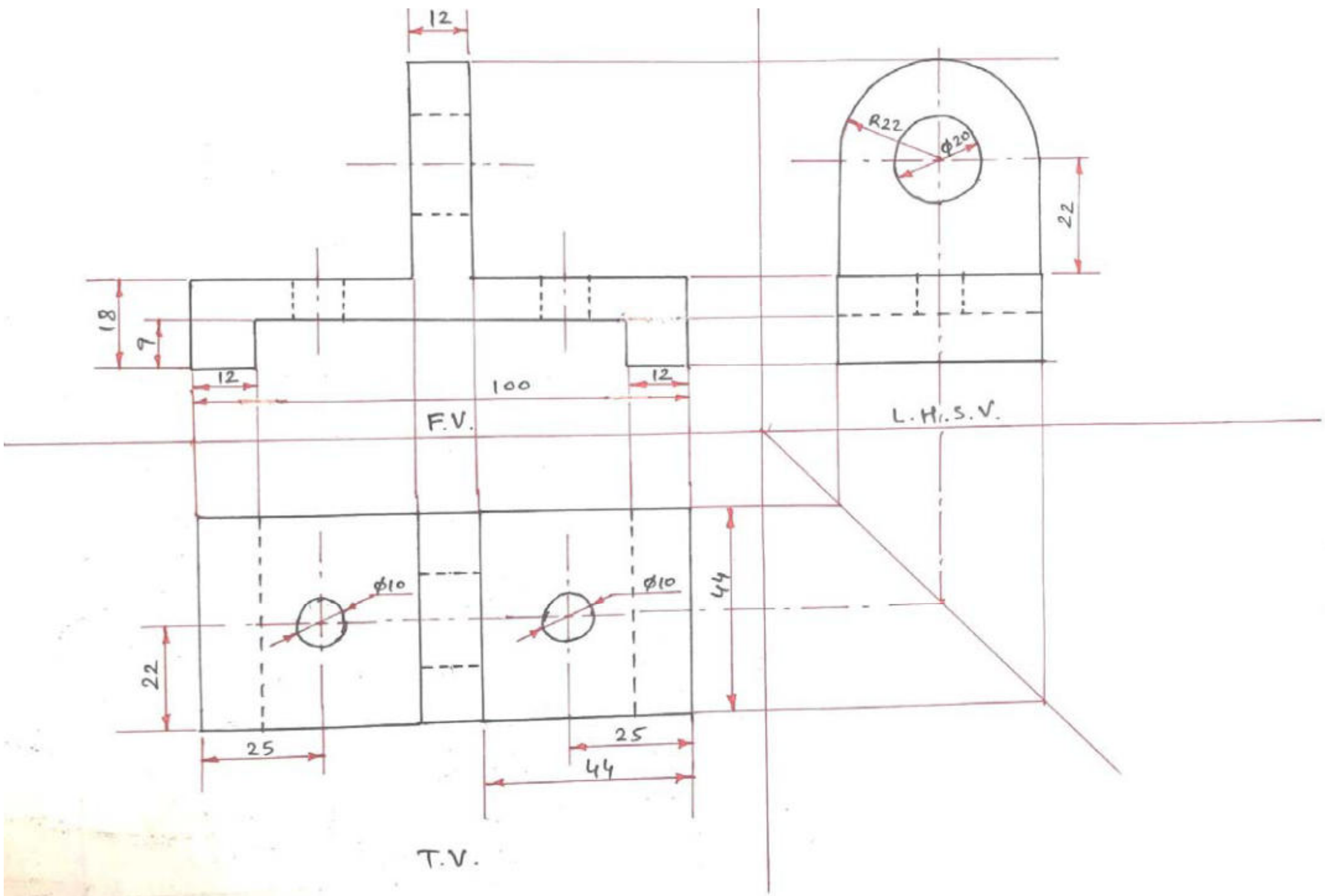
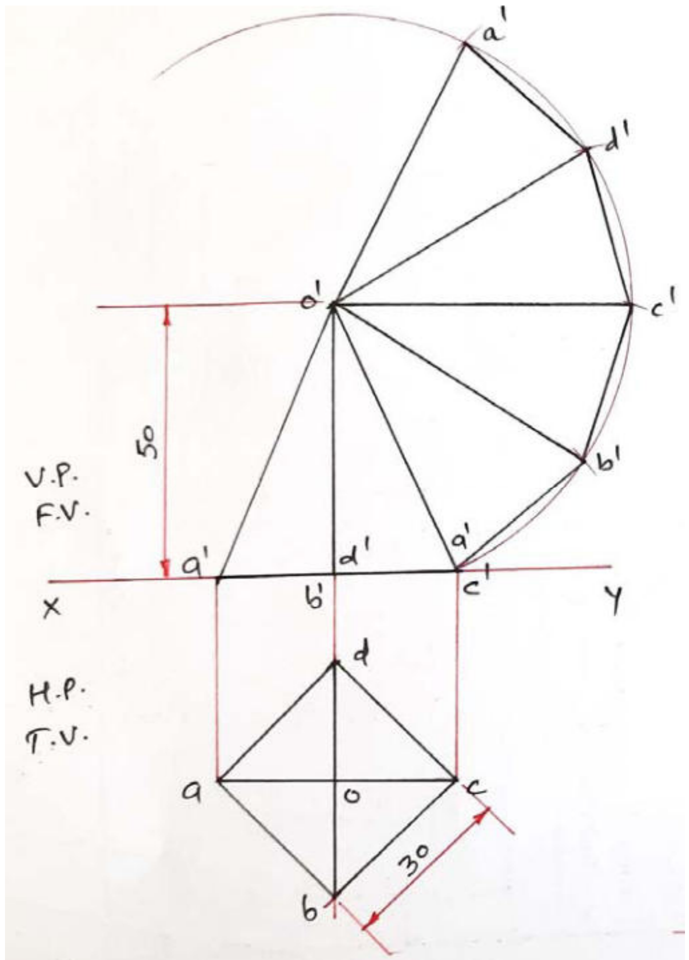



Figure 2

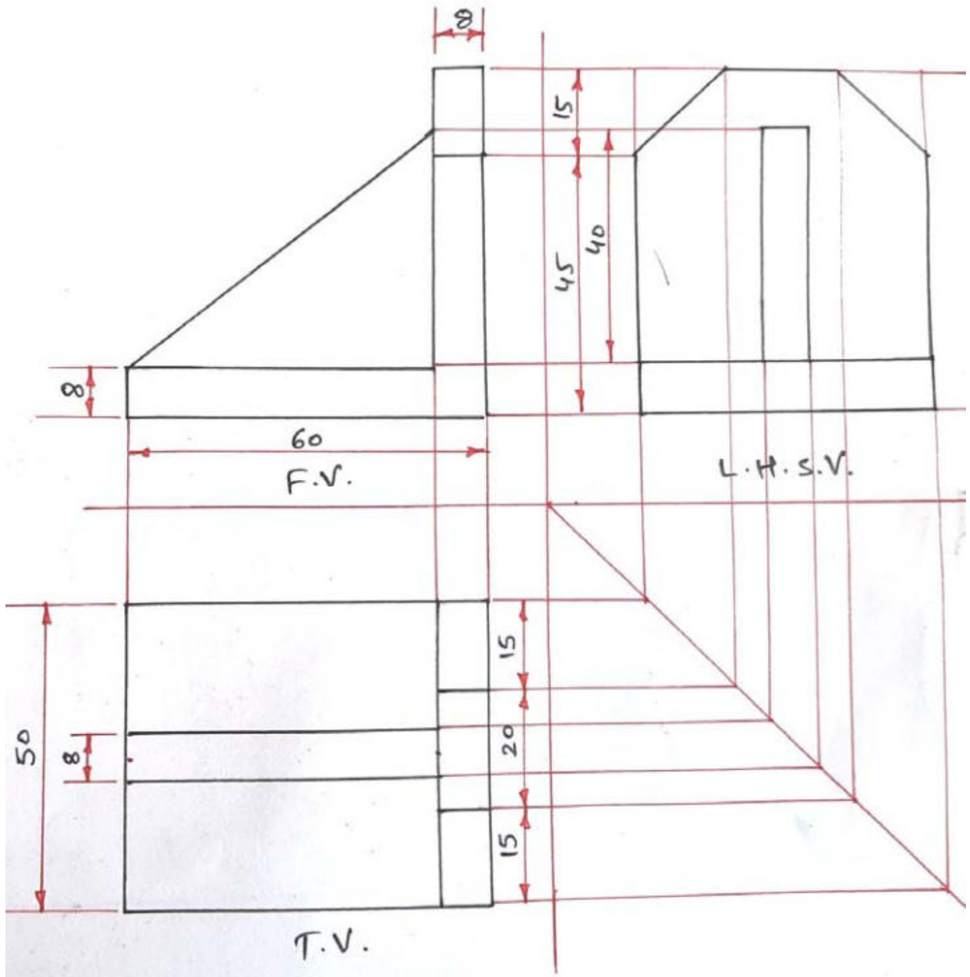


**Q.4 (a) OR** Draw the development of the lateral surface of a square pyramid, side of base 30 mm and height 50 mm, resting with its base on H.P. All edge of the base are equally inclined to V.P. **( 03 Marks)**



**Q.4 (b) OR** Draw front view of an object shown in figure 3 using first angle projection methods. **( 04 Marks)**

**Q. 4 (c) OR** Draw top view and left hand side view of object shown in figure 3 using first angle projection methods. **( 07 Marks)**



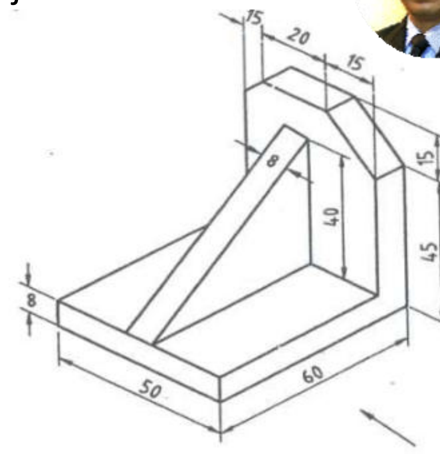




Figure 3

**Q.5 (a)** Explain any three AutoCAD commands in short (1) Spline (2) Rotate (3) Polygon (4) Multiline **( 03 Marks)**

1. **Spline command**

Spline command is used to draw a smooth curve passing through number of given points.

Spline command can be given in three different ways as under.

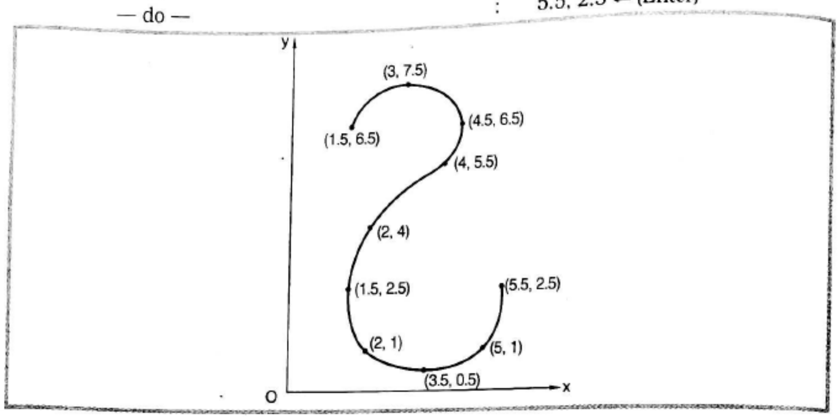
(1) Click by mouse on 'SPLINE' toolbar icon.

(2) Type by keyboard word 'SPLINE' in the command window.

(3) Select 'SPLINE' option from draw menu.

**Problem : Draw a curve (2 ↵) passing through given points (1.5, 6.5), (3, 7.5), (4.5, 6.5), (4, 5.5) (2, 4), (1.5, 2.5), (2, 1), (3.5, 0.5), (5, 1) and (5.5, 2.5).**

| | | |
|---|---|---|
| Command | : | Spline ↵ (Enter) |
| Specify first point or [object] | : | 1.5, 6.5 ↵ (Enter) |
| Specify next point | : | 3, 7.5 ↵ (Enter) |
| Specify next point or [close, fit tolerance] (Start Tangent) | : | 4.5, 6.5 ↵ (Enter) |
| — do — | : | 4, 5.5 ↵ (Enter) |
| — do — | : | 2, 4 ↵ (Enter) |
| — do — | : | 1.5, 2.5 ↵ (Enter) |
| — do — | : | 2, 1 ↵ (Enter) |
| — do — | : | 3.5, 0.5 ↵ (Enter) |
| — do — | : | 5, 1 ↵ (Enter) |
| — do — | : | 5.5, 2.5 ↵ (Enter) |




2. Rotate

Rotate command is used to rotate object at definite angle.

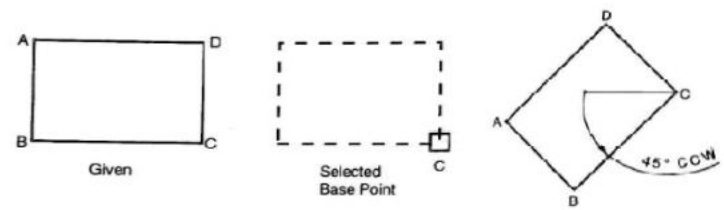
Example: rotate rectangle at 45°

Command: rotate

Select object: select object by mouse (enter)

Specify base point: specify base point through which object is to be rotated (enter)

Specify angle of rotation: 45 (enter)



**Q.5 (a)** Explain any three AutoCAD commands in short (1) Spline (2) Rotate (3) Polygon (4) Multiline **( 03 Marks)**

3. Polygon

Polygon command is used to draw any regular shape polygon

Example: Trianlge, square, pentagon, hexagon etc.

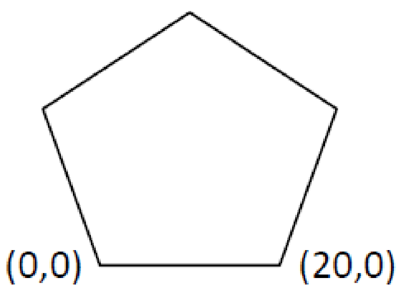
Explain steps to draw Pentagon of 20 mm size.

Command: Polygon

Number of side: 5 (Enter)

Specify first end point of edge: 0,0

Specify second end point of edge: 20,0




4.

**Multiline Command**

This command is used to draw parallel lines with specified offset distance (scale) between two lines.

Multiline command can be given in the following two ways.

(1) Type by keyboard word 'MLINE' in the command window.

(2) By selecting multiline option from the draw menu.

Command : MLINE ↵ (Enter)

Current Settings : Justification = Top, Scale = 1.00

Style = Standard

Specify start point or [Justification/Scale/Style] : 5.0, 10.0

Specify next point : 20.0, 15.0 See Fig. 18

Justification is the alignment of two lines w.r.t. reference Top/Zero/Bottom. [Shown in Figure]

Scale is the spacing between two parallel lines. [Shown in figure-19]

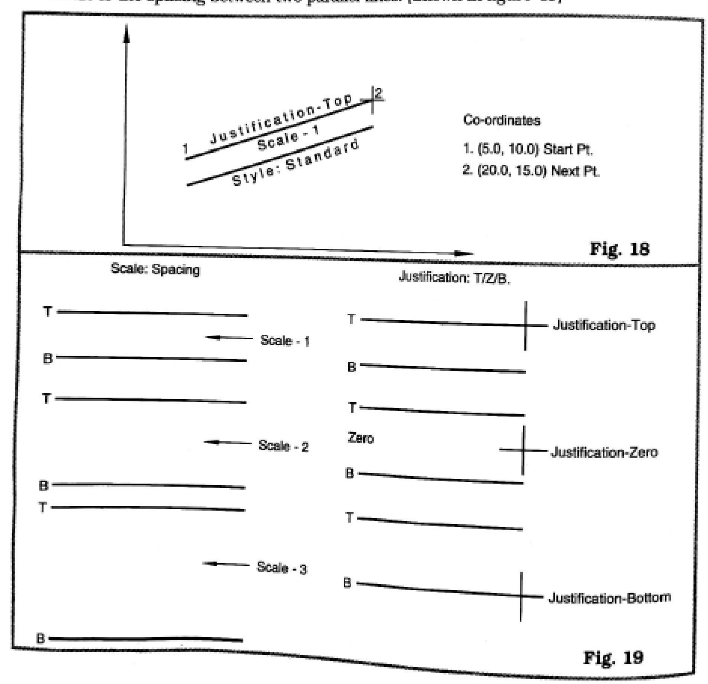
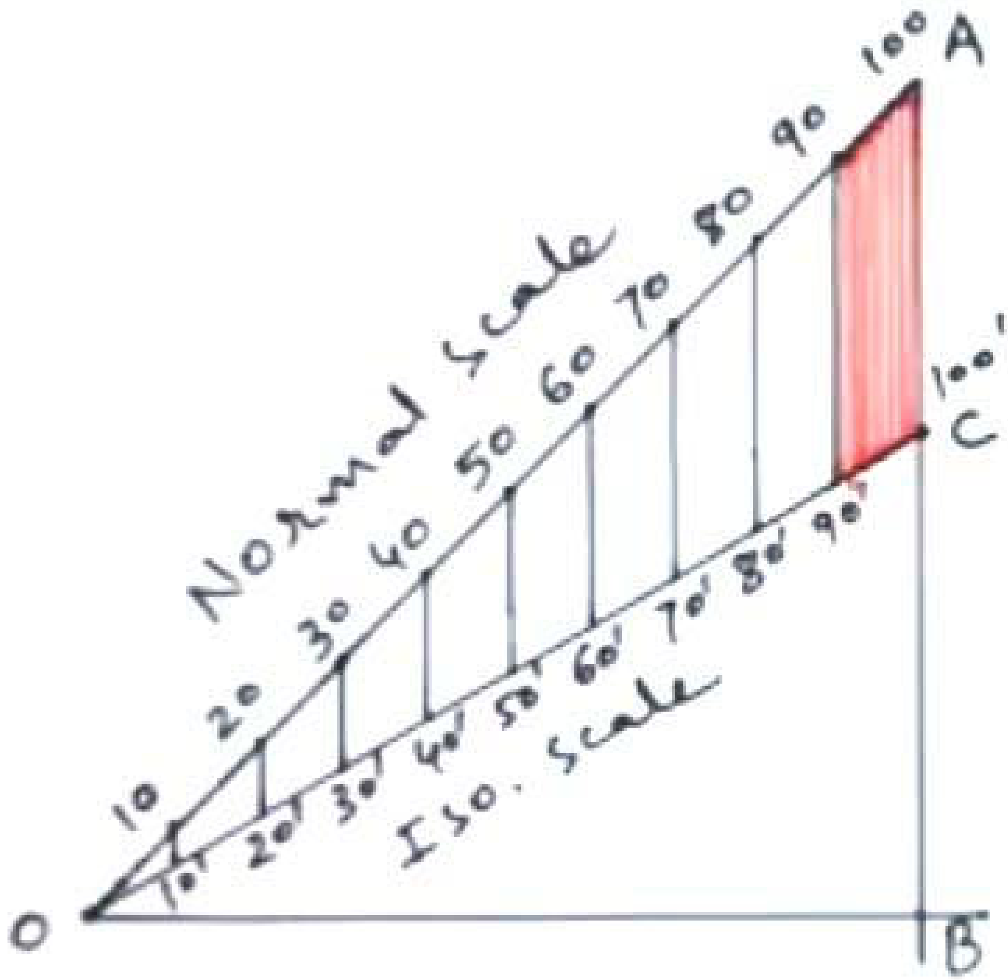


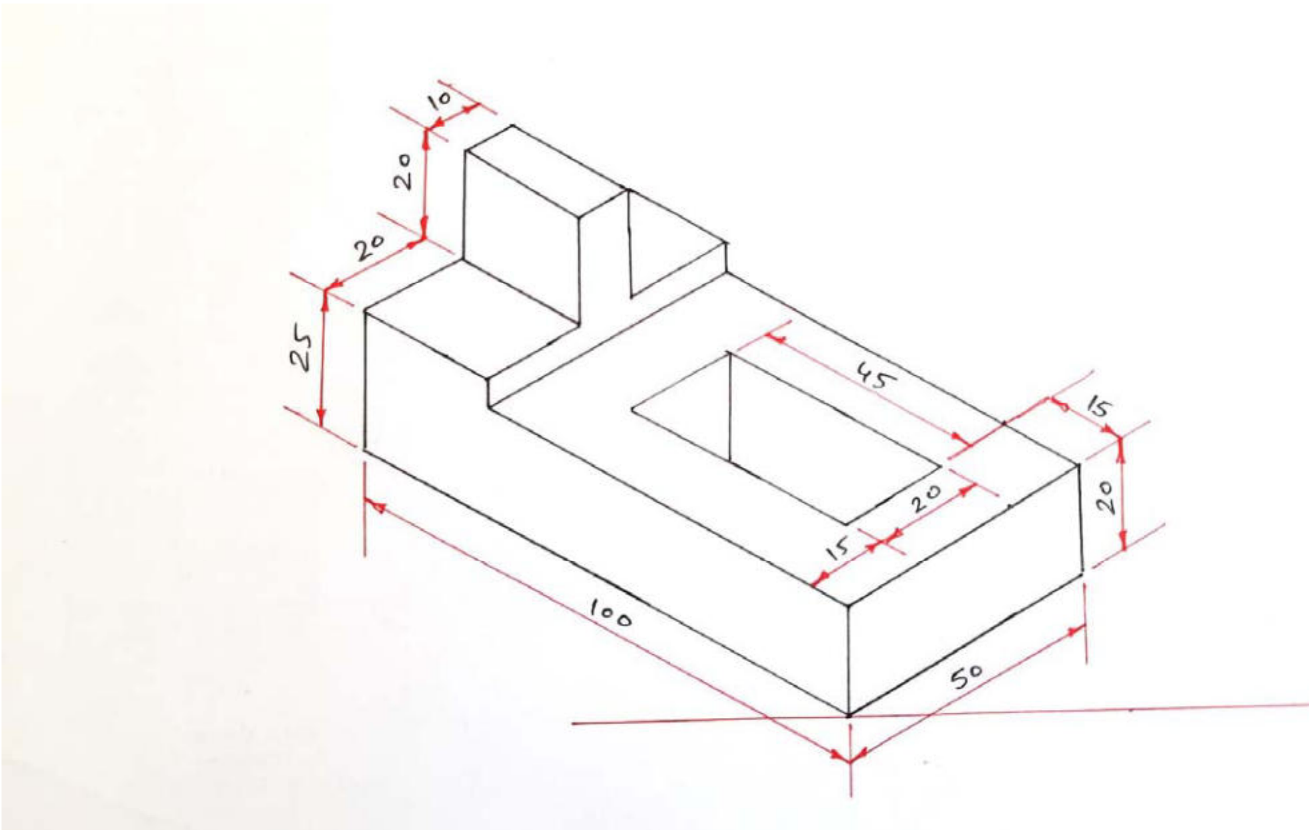

Fig. 18

Fig. 19

**Q.5 (b)** Draw isometric scale showing 100 mm isometric length equivalent to 100mm normal length. **( 04 Marks)**



**Q.5 (c)** Draw isometric projection of an object whose front view and top view are shown in figure 4. **( 07 Marks)**



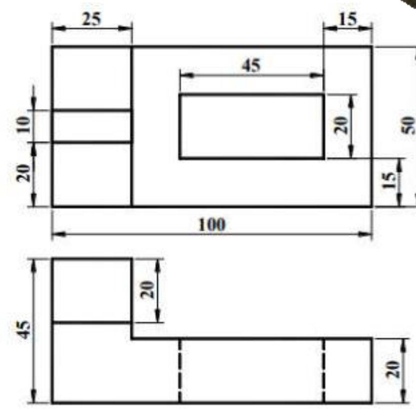




Figure 4

**Q.5 (a) OR** Explain any three AutoCAD commands in short. (1) Circle (2) Array (3) Mirror (4) Filet Command **( 03 Marks)**

## 1. Circle

Following methods is used to draw circle.

1. Center point, radius method
2. Center point, diameter method
3. 2 point method
4. 3 point method
5. Tangent, tangent, radius (TTR) method.

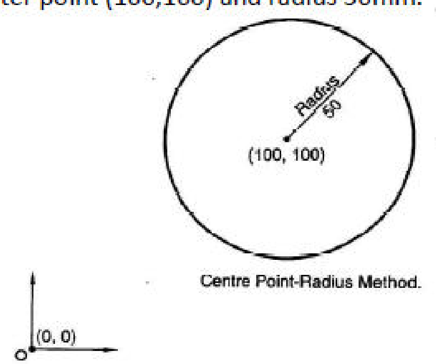
Method 1: Center point, radius method

Command: Circle

3P/2P/TTR/<center point>: 100,100 {Center point method is default method by AutoCAD}

Diameter / <Radius>: 50 {Radius parameter is default parameter by AutoCAD}

Circle will be drawn having center point (100,100) and radius 50mm.




Centre Point-Radius Method.

## 2. Array

Array command is used to create multiple copies of selected object in rectangular or polar/circular pattern.

Pattern 1 : Rectangular

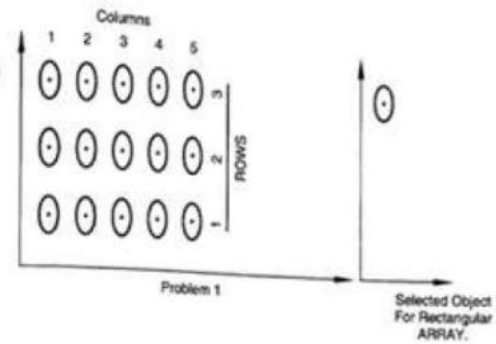
Command : Array

Enter the type of array [Rectangular / Polar]: R (enter)

Enter number of raw: 3 (enter)

Enter number of column: 5 (enter)

Specify distance between the column: 3 (enter)




Pattern 2: Polar / Circular

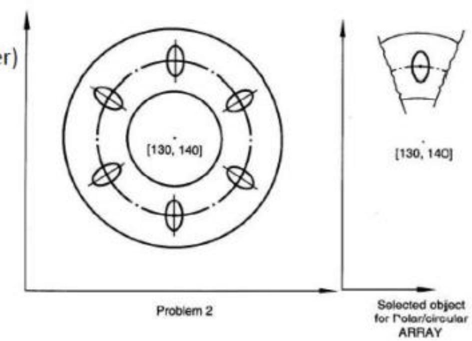
Command : Array

Enter the type of array [Rectangular / Polar]: P (enter)

Locate center point of array: 130,140 (enter)

Specify number of item in the array: 6 (enter)

Specify angle: 360 (enter)



**Q.5 (a) OR** Explain any three AutoCAD commands in short. (1) Circle (2) Array (3) Mirror (4) Fillet Command **( 03 Marks)**

3. Mirror

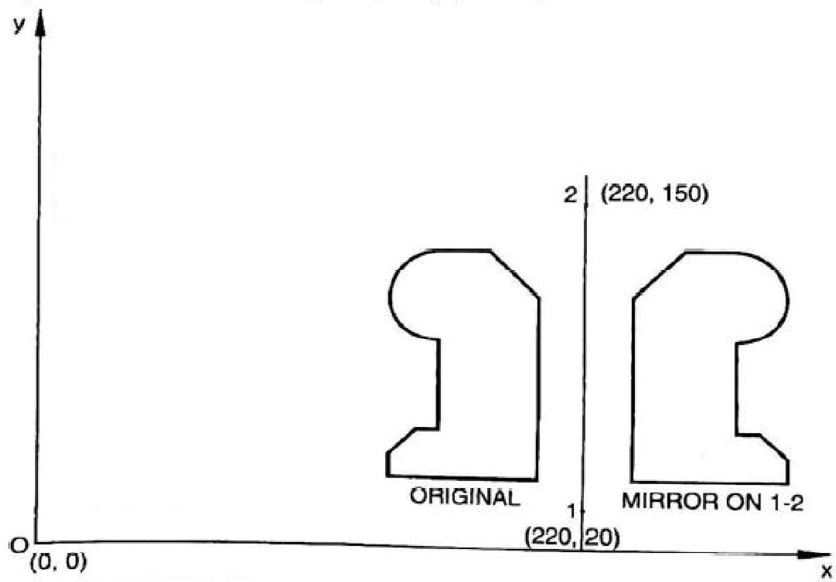
Mirror command is used to obtain mirror image of an object.

Command : Mirror

Select object: select object which is to be converted into mirror image

Specify first end point of mirror line: (220, 20) (enter)

Specify second end point of mirror line: (220,150) (enter)




4. Fillet

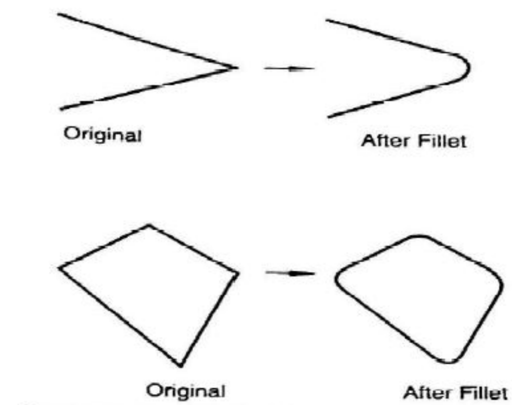
Fillet command is used to round off the sharp corners.

Command: Fillet

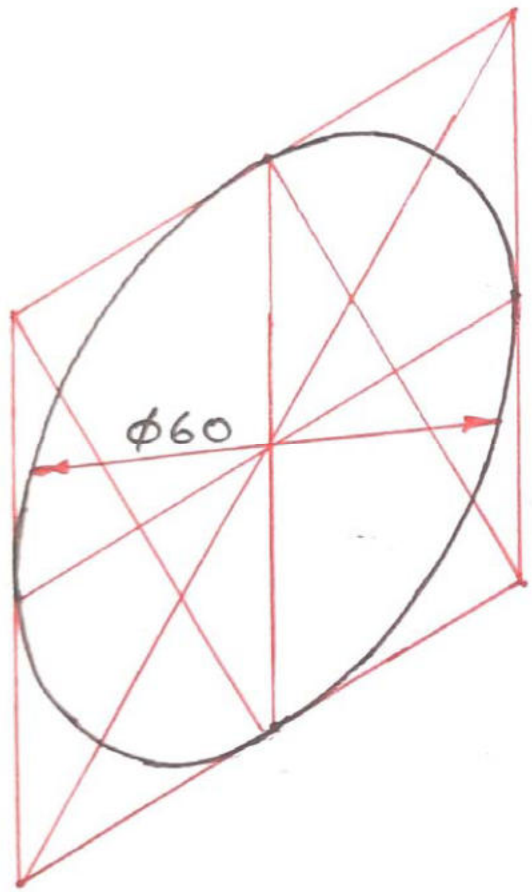
Specify fillet radius: 15 (enter)

Select first object: select line 1 (enter)

Select second object: select line 2 (enter)



**Q. 5 (b) OR** Draw isometric shape of a circle having 60 mm in diameter using four centre methods. **( 04 Marks)**



**Q. 5 (c) OR** Draw isometric projection of an object whose front view, top view and side view are shown in figure 5.
**( 07 Marks)**

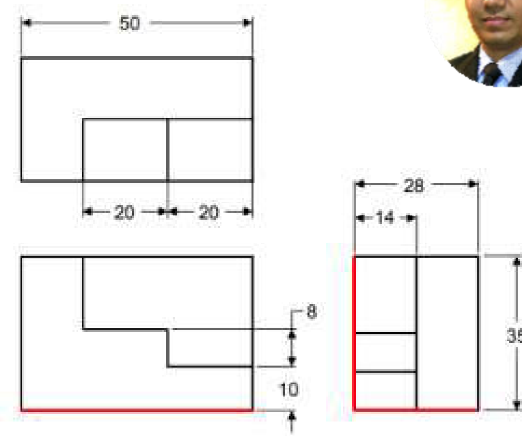



Figure 5